COMPLETE PROJECT DOCUMENT • SOLANA NETWORK

TLC Protocol: General Overview & Executive Summary

Integrated On-Chain Mortgage Lending & Residential Access Ecosystem

1. PLAIN ENGLISH PROJECT OVERVIEW

Imagine you want to **buy or finance a home**, but traditional banks require endless paperwork and charge steep interest rates. On the other hand, people with savings earn almost nothing in bank accounts, while in the crypto world, investors are searching for real-world assets (RWAs) to back their capital with tangible value.

The **TLC Protocol** bridges these two worlds using Solana blockchain technology:

- **For Borrowers:** Property owners place their real estate into a *legal trust* (an armored legal entity) and receive fast financing provided by lenders worldwide.
- **For Lenders (Liquidity Providers):** Investors deposit capital into a shared pool and earn real yield in digital dollars (USDC or BTC), backed directly by physical properties. If they need liquidity before loan maturity (1–5 years), they can sell their position token on secondary markets.
- **Protection & Opportunity (Collateral & TLC Token):** Properties are over-collateralized at 200% of the loan amount. If a borrower defaults, a 7-day fast-track auction is triggered where **priority to purchase the property at a discount goes to long-term holders of the TLC token**.

2. INSTITUTIONAL PROTOCOL OVERVIEW

At a technical level, TLC Protocol is an on-chain lending engine built on Solana that is backed by Real-World Assets (RWA). By structuring real estate collateral through legally binding bankruptcy-remote trusts (SPVs), the protocol delivers a compliant pathway for crypto capital into over-collateralized mortgage debt. In default scenarios, smart-contract execution triggers an expedited auction mechanism that preserves LP liquidity while driving structural utility to the native token.

3. ECOSYSTEM VALUE PROPOSITIONS

🏠 Real Estate Borrowers

Access non-bank, competitive long-term liquidity (1–5 years) collateralized by property assets without traditional banking friction.

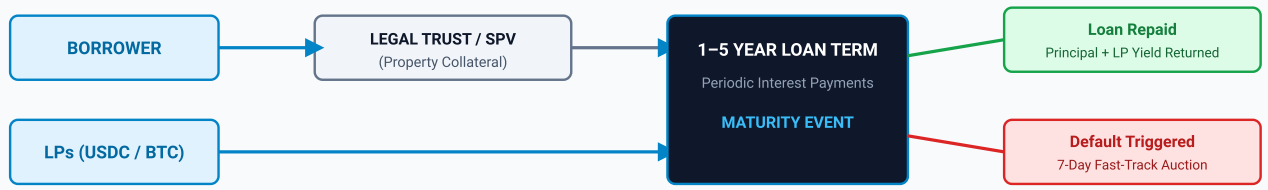
💎 Liquidity Providers (LPs)

Earn Real Yield paid in stablecoins (USDC/BTC) backed by real physical housing with a 50% max LTV safety buffer.

🪙 TLC Token Holders

Gain exclusive priority rights to buy foreclosure properties at a discount during the 7-day grace period based on holding duration.

4. PROTOCOL ARCHITECTURE & OPERATIONAL FLOW



A. Origination & Collateral (Years 1–5)

1. **Legal Structure:** Property deed locked in legal trust / SPV.
2. **Conservative LTV:** Max 50%–60% LTV based on certified appraisal.
3. **Secondary Liquidity:** tUSDC debt position tokens issued for DEX trading.

B. Fast-Track Auction Phases (7 Days)

1. **Day 0:** Smart Contract opens 168-hour auction window on default.
2. **Days 1–3:** First priority for TLC holders based on holding duration score.
3. **Days 4–5:** Second priority opens to loan pool LPs.
4. **Days 6–8+:** Open public bidding for DAO community & institutions.

5. RISK ANALYSIS & COMPETITIVE ADVANTAGES

Dimension	Traditional Challenge	TLC Protocol Solution
Liquidation Speed	Traditional judicial foreclosures require 12 to 36 months to settle.	**7-Day On-Chain Auction:** Programmatic settlement enforced through the legal trust.
Anti-Whale Protection	Institutional funds swooping in to grab discounted housing ahead of retail.	**Time-Weighted Score:** Purchase priority is governed by duration of TLC token holding.
Collateral Safety	Real estate market value fluctuations over loan lifespans.	**50% Max LTV & Re-Appraisals:** Generous equity buffer ensuring 100% LP principal recovery.
LP Capital Flexibility	Capital locked up in real estate loans for up to 5 years without exit.	**Liquid Debt Tokens:** Freedom to exit position by selling debt tokens on DEXs.

6. SOLANA TECHNOLOGICAL STACK

Smart Contracts

Built with Anchor (Rust) for high throughput and verifiable security.

Token-2022

Utilizes Transfer Hooks to track holding duration on-chain immutably.

Legal Oracle

Multi-signature linkage synchronizing smart contracts with the legal trust.

7. ECONOMIC INCENTIVE MODEL (FLYWHEEL)

🎯 Structural Demand

To secure priority access for buying discounted housing, users acquire and hold TLC tokens, locking up circulating supply.

🔥 Deflationary Engine

Protocol fees from origination and auctions fund token buybacks, strengthening market liquidity.

📈 Sustainable Real Yield

Yield is generated from real-world mortgage payments rather than token inflation, ensuring long-term financial viability.